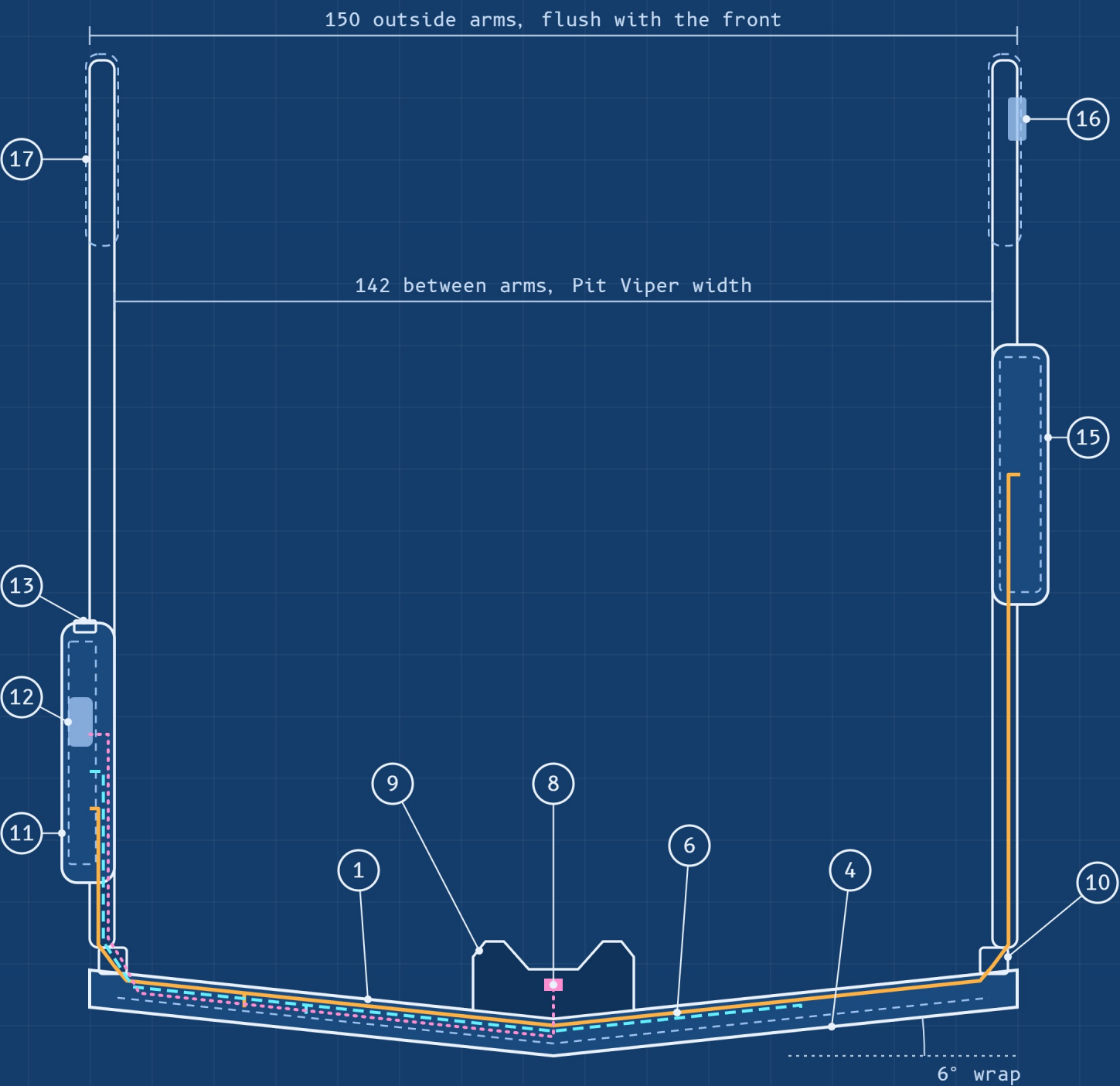
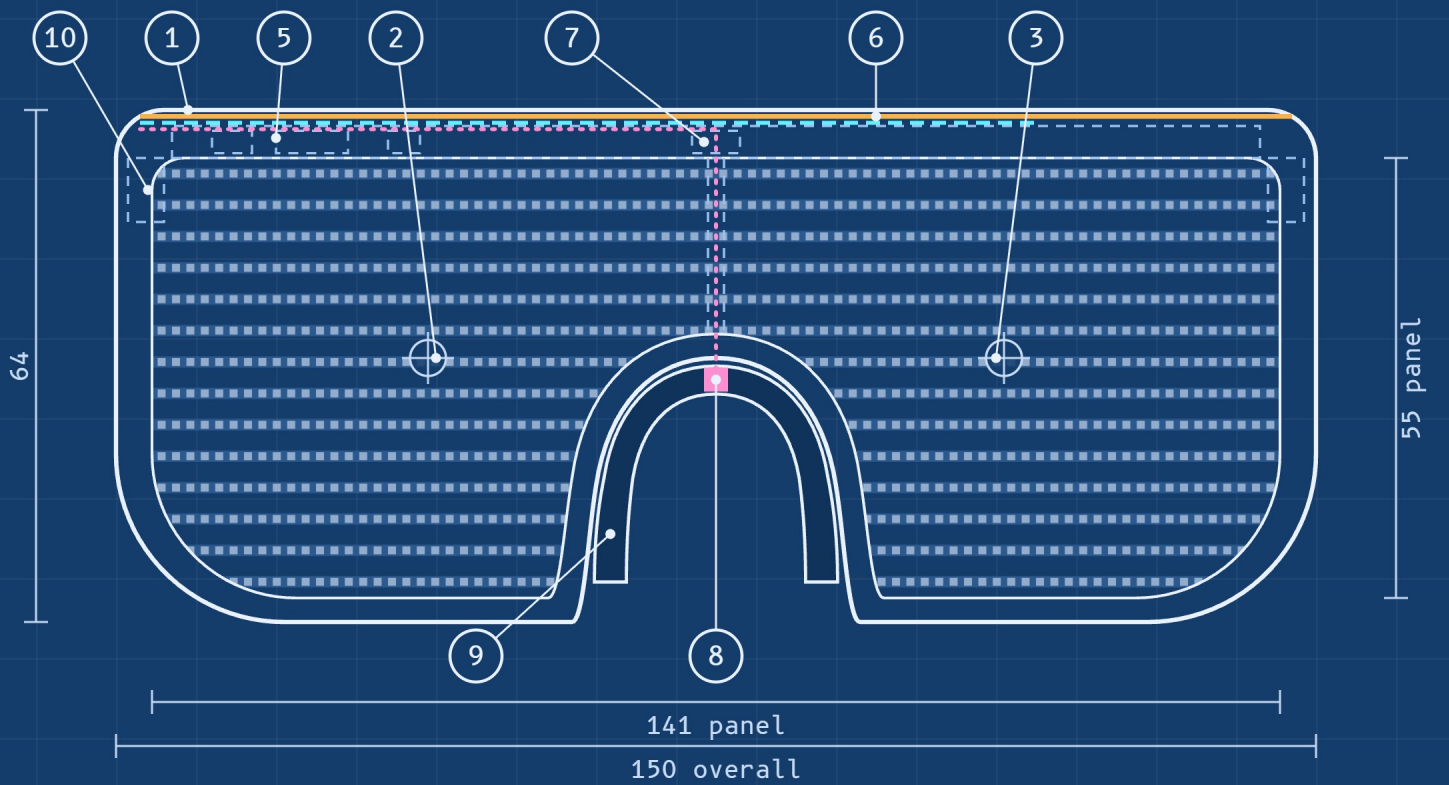


CAPTION SHADES / GENERAL ARRANGEMENT



TOP VIEW · LOOKING DOWN, FRONT AT BOTTOM

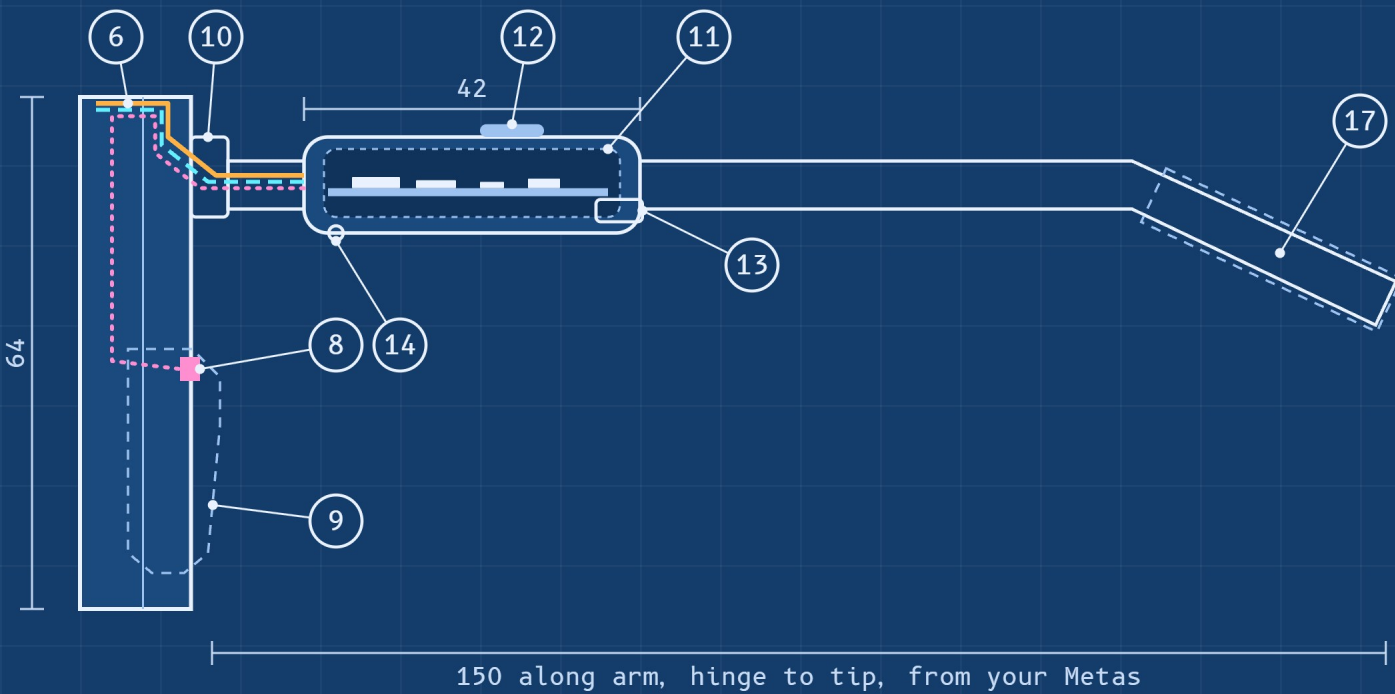


FRONT VIEW · FACING THE WEARER

Width and nose opening from your Pit Viper Original Wides: arms flush with the 150 front, 142 between them; nose 36 × 33.
From your Ray-Ban Metas [50□22 150]: 150 arms, eye centres [+] 72 apart. All in mm, nominal.
Panel: 14 bars, 2.5 mm gaps, about 64% see-through, about 930 LEDs. About 45 g with battery; your Metas are about 49 g.

NO.	PART	NOTES
1	Front frame	Printed nylon [PA12], matte black. Full rim around both panels.
2	LED panel, right	0.8 mm four-layer board, black. 14 bars, up to 39 LEDs per bar.
3	LED panel, left	Mirror of 2. Meets it at the centre seam; each is angled 6° back.
4	Front window	0.75 mm polycarbonate, clear or light smoke. Protects the LEDs.
5	Component strip	Power switch, 5 V boost, signal booster, hidden under the top rim.
6	Wiring channel	Runs inside the top rim, hinge to hinge.
7	Centre jumper	5 V, ground and left-panel data across the seam.
8	Voice sensor	In the top of the saddle, on the bridge of the nose. Wire rises inside the centre rib.
9	Nose saddle	One-piece silicone, Pit Viper style. Snaps onto the frame; swap sizes for fit.
10	Spring hinges [2]	Hollow barrel. A flex cable passes through with a service loop.
11	Right arm, driver pod	nRF52840 Bluetooth board with battery charger and motion sensor.
12	Button	Mute and power.
13	USB-C	Charging and firmware.
14	Mic port	Air microphone that supplies the words.
15	Left arm, battery pod	LiPo pouch cell, about 250 mAh, 4 × 12 × 40 mm.
16	Tether contacts	Optional magnetic link to a pocket battery for full days.
17	Arm tips [2]	Silicone sleeves.

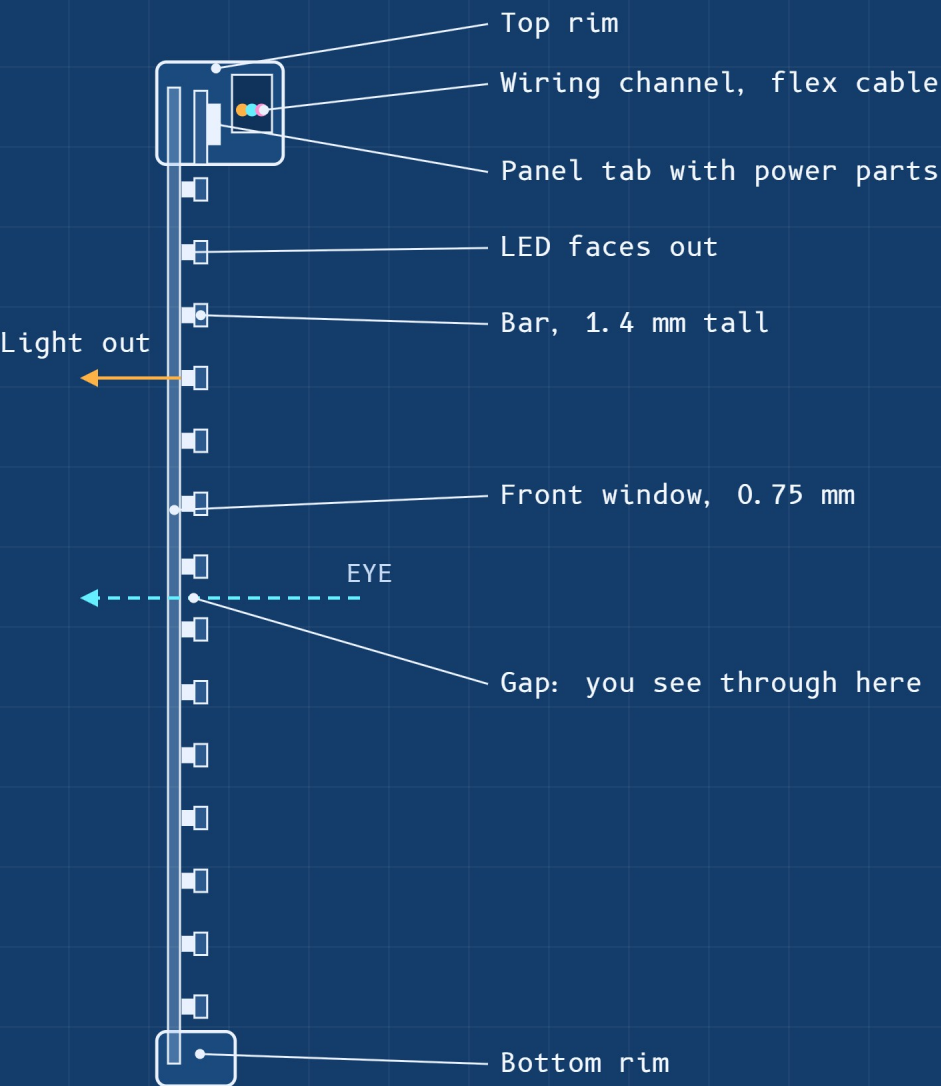
— Power - - - - Data to the panels ······ Voice sensor - - - - Hidden part



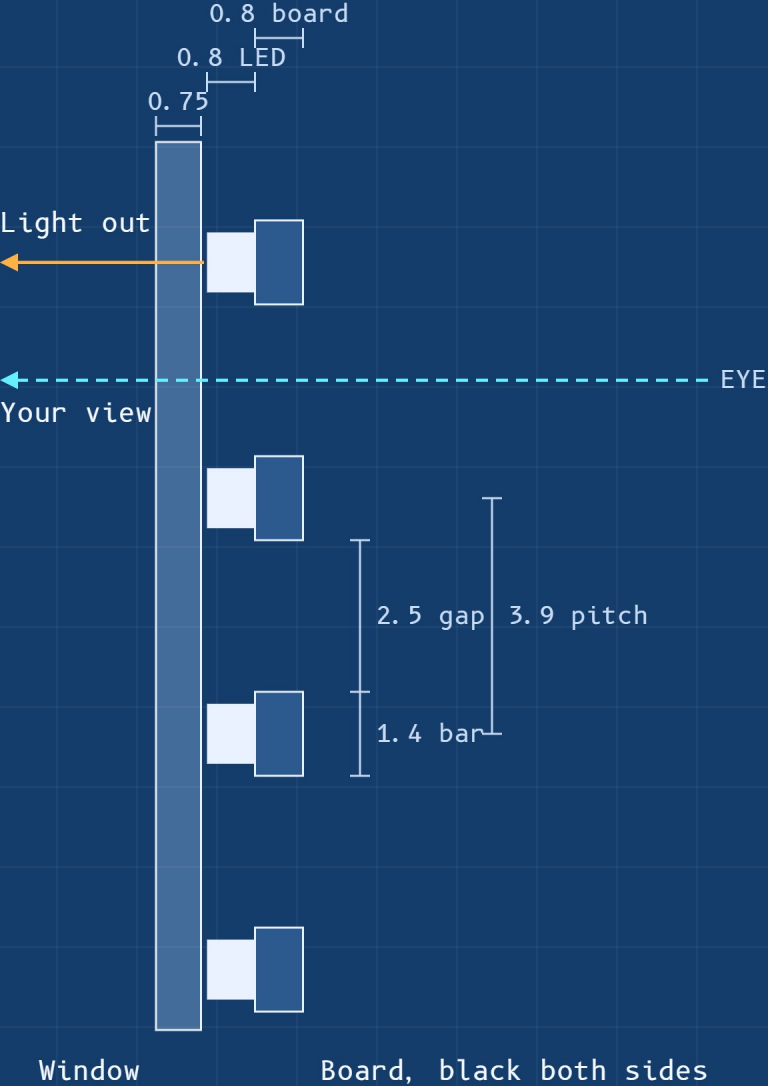
RIGHT SIDE VIEW · DRIVER POD CUT AWAY

CAPTION SHADES / CUTAWAYS

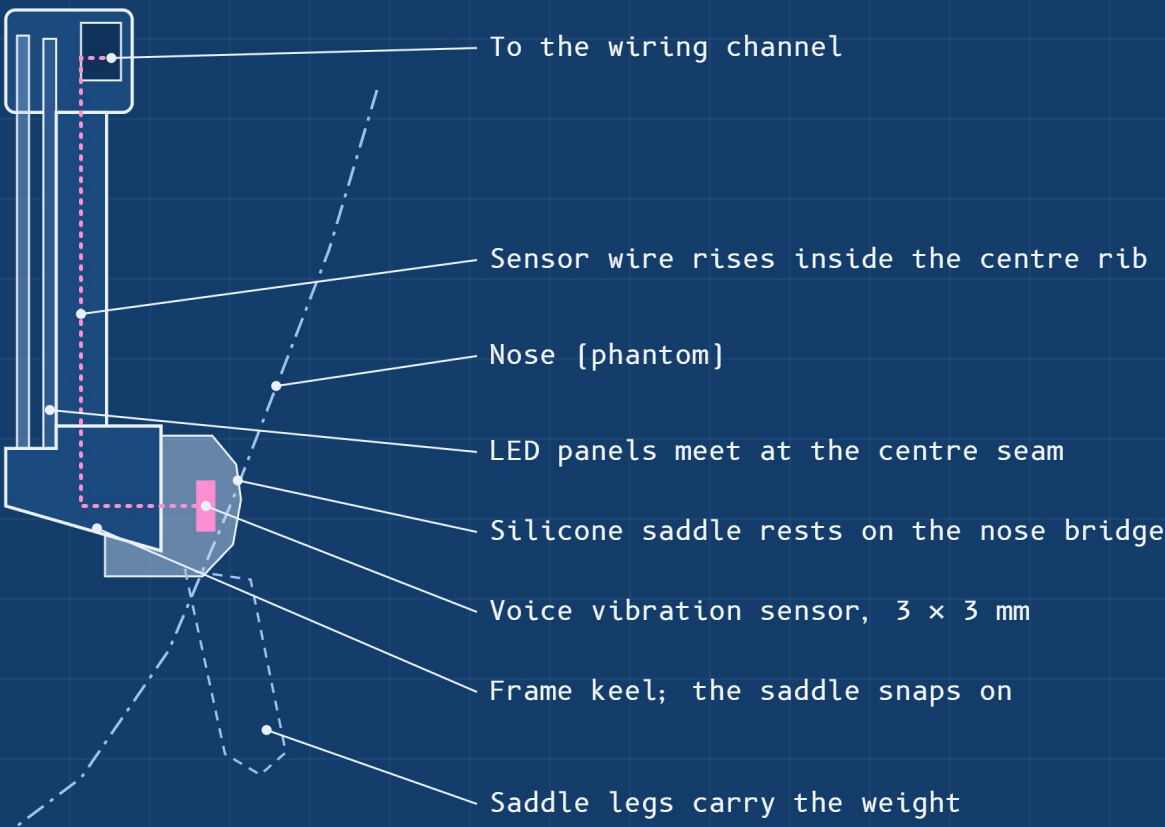
SECTION A-A · THROUGH THE LEFT LENS



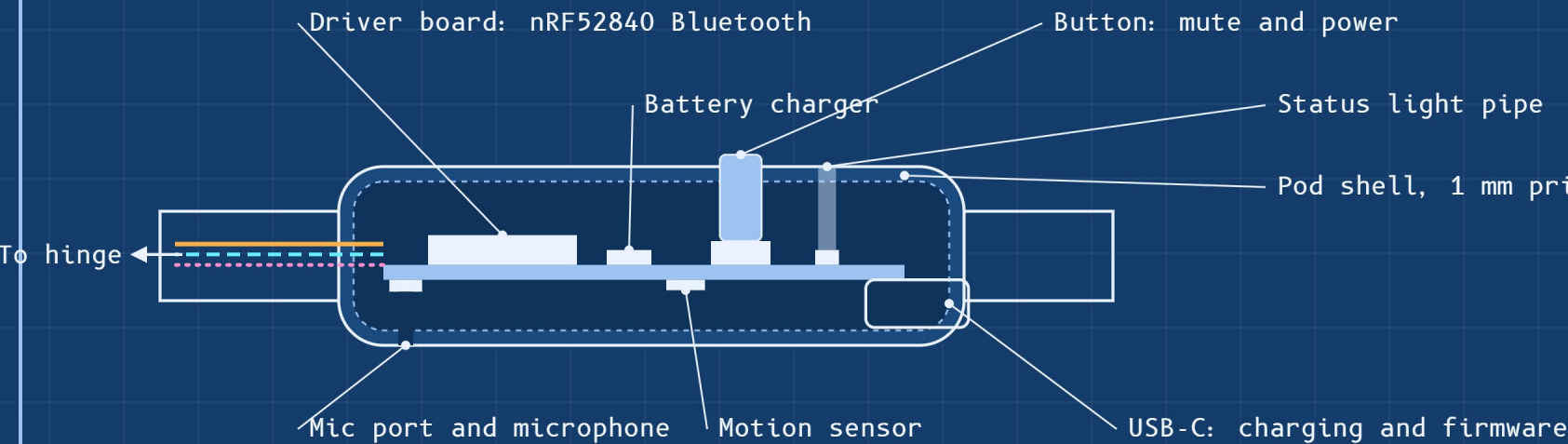
DETAIL A1 · THREE BARS, ENLARGED



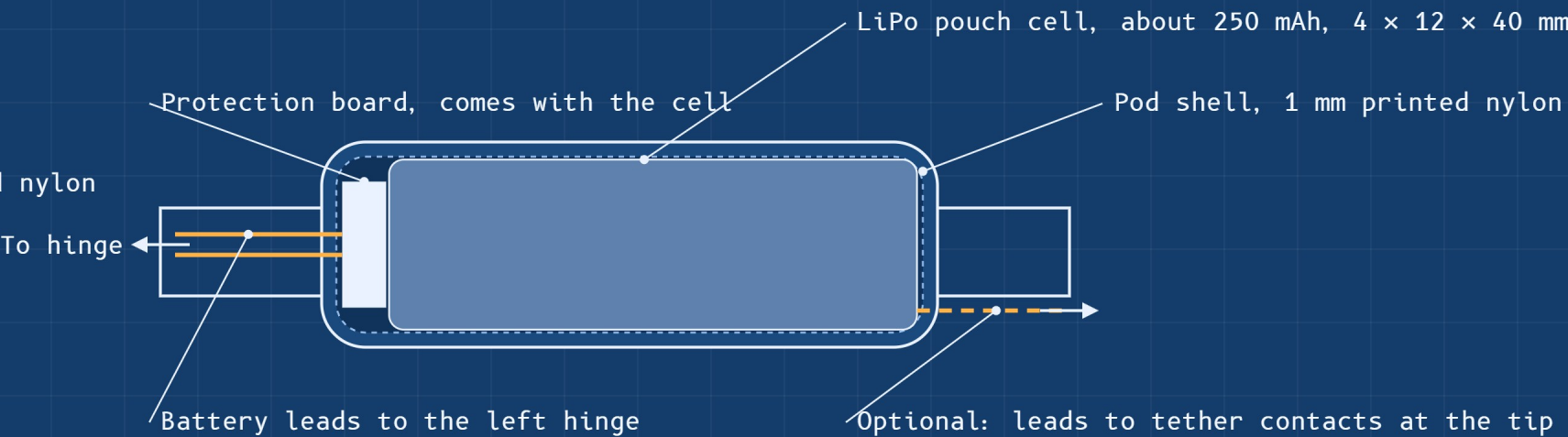
DETAIL D · BRIDGE SENSOR, CENTRE SECTION



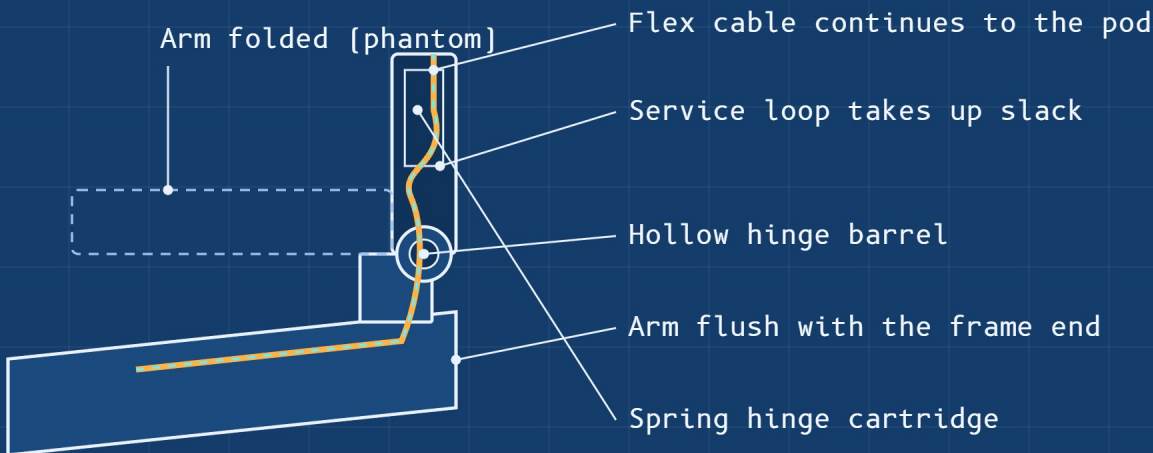
DETAIL B · DRIVER POD, RIGHT ARM, SHELL CUT AWAY



DETAIL C · BATTERY POD, LEFT ARM, SHELL CUT AWAY



DETAIL E · HINGE, TOP VIEW



WIRES · THROUGH EACH JOINT

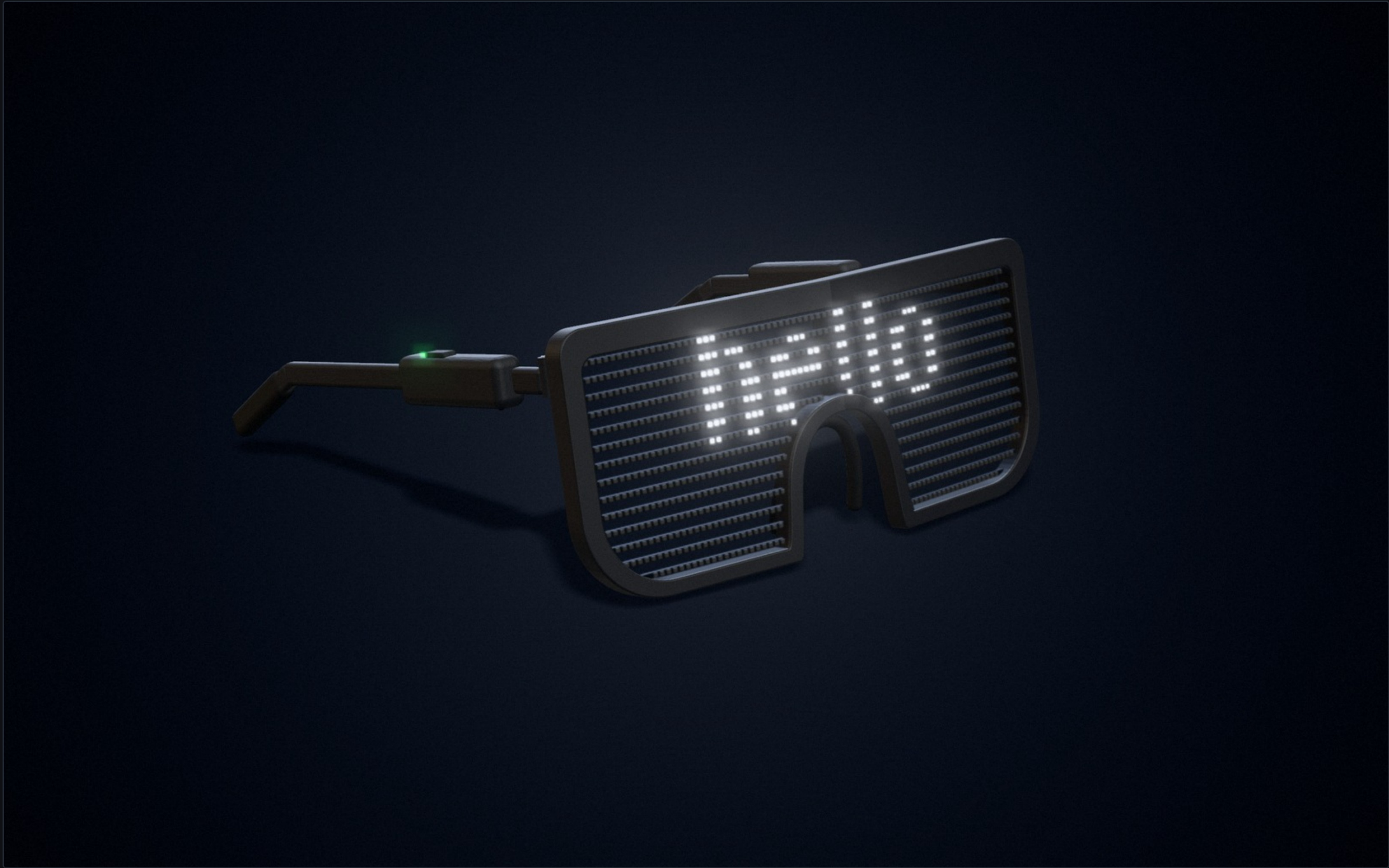
Right hinge	8	Battery +, ground, LED power on/off, left panel data, right panel data, sensor data, sensor clock, sensor wake
Left hinge	3	Battery +, ground, battery temperature
Centre jumper	3	5 V, ground, left panel data

Dimensions in mm, nominal. The battery in the arm is a buffer; for a full day, plan on the tether.

WEIGHT · ESTIMATE

Frame and arms, printed nylon	17 g
LED panels [2] with LEDs	7 g
Front window	6.5 g
Battery	5.5 g
Driver board	3 g
Hinges, nose saddle, screws	4 g
Sensor, mic, wiring, power parts	2 g
Total, about	45 g

CAPTION SHADES / RENDERS



CALM Everyday talking: white serif captions in the band above the nose.



SHOUT Louder words go bold and rainbow; lightning fills the full-height sides.



BACK Driver pod with button, status light and USB-C; battery pod on the far arm; nose saddle.



ON YOU Composite on your photo. The window is shown in dark smoke, since your eyes were behind a mirror lens.

Built from the Rev D blueprint: 150 mm front with the arms flush at the ends, Pit Viper-size nose opening [36 × 33], 150 mm arms, 6° wrap, 1 mm LEDs on black bars, light-smoke window, matte black printed nylon, silicone saddle and tips.

CAPTION SHADES	SHEET 3 OF 3 Renders	REV D • 9 OCT 2026 RENDERED BY CLAUDE
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